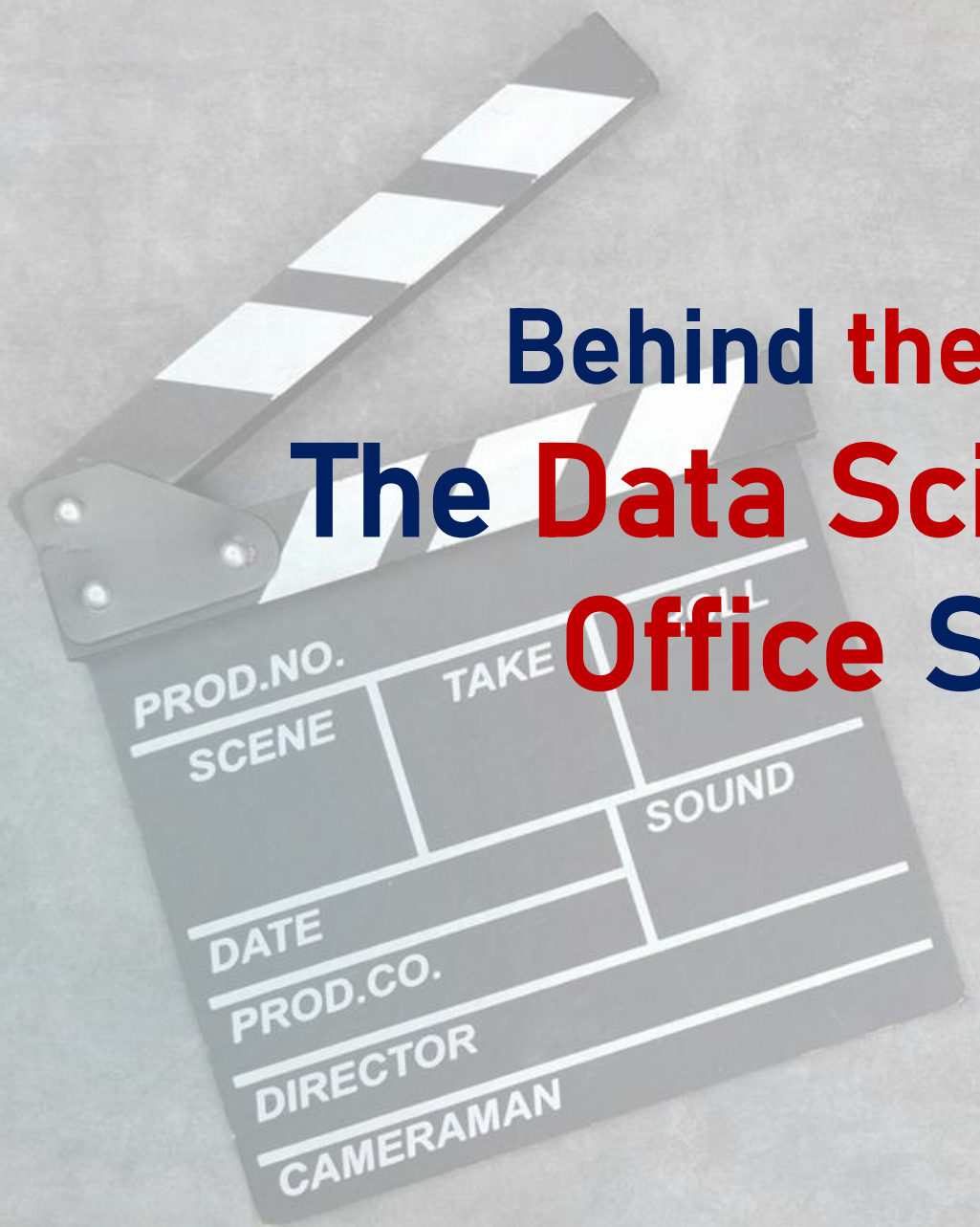




Behind the Screens : The Data Science of Box Office Success

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Agenda



- ▶ Objective & Dataset
- ▶ Data Analysis Workflow (5 Stages)
 - ▶ Data Cleaning
 - ▶ Data Extraction (OMDb API)
 - ▶ Statistical Analysis
 - ▶ Regressor Model Generation
 - ▶ Model Evaluation
- ▶ Insights
 - ▶ Budget Outliers & Gross
 - ▶ R-Rated Films in Outliers
 - ▶ ROI by Genre & Rating
 - ▶ Studio & Genre ROI Highlights
 - ▶ Star-Writer Duos Analysis
- ▶ Strategic Takeaways & Actions

Objective & Dataset



- ▶ **Main Objective:** Use data-driven analysis to uncover the factors—like budget size, genre, rating, and studio—that most influence a movie's financial efficiency (ROI), and provide actionable insights for studios to make smarter investment decisions.
- ▶ The source dataset is taken from Kaggle with link below
- ▶ [Kaggle Movie Dataset](#)
- ▶ **Goal:** Understand what leads to profitable movies

▶ Key metrics:

Topmost Gross
2.85 Billion

Topmost Budget
356 Million

**Top company based on
ROI**
Marvel Studios - 4.65

**Top Genre based on
profit**
Action - 156 Billion

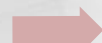
Data Analysis Workflow



Data Cleaning
With Pandas



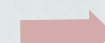
Data Extraction
OMDb API



Statistical Analysis
Describe()



Model-Training
RadomForest-
Regressor



Model Evaluation
R2_score,
gross_mean error

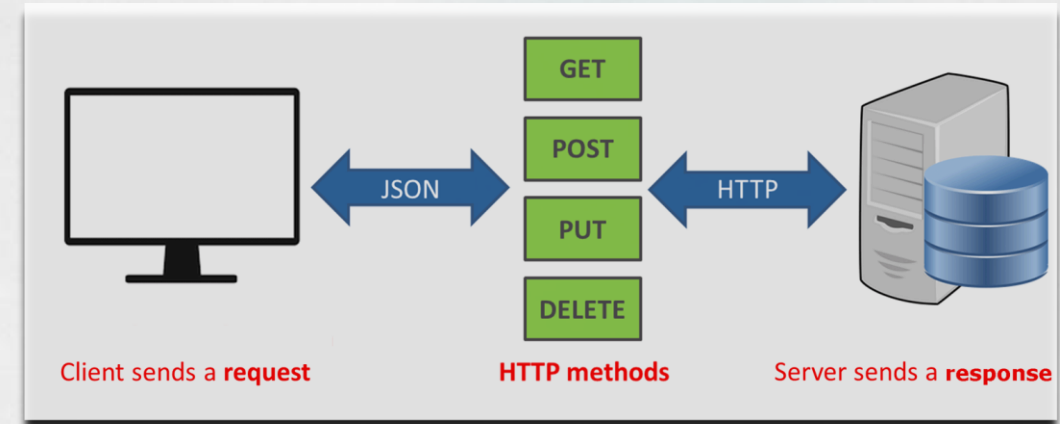
Stage 1 – Data Cleaning

- ▶ Tools:
 - ▶ Pandas – For data loading and transformation,
 - ▶ NumPy – For numerical operations
- ▶ Actions:
 - ▶ Read CSV → DataFrame
 - ▶ Fixed data types
 - ▶ Standardized release date -> Converted to DateTime
 - ▶ Created new 'released year' column using dt.year
 - ▶ Merged Unrated and Not Rated column -> Not Rated
 - ▶ Dropped redundant columns

“Building a clean foundation to unlock accurate movie insights.”

Stage 2 – Data Enrichment via OMDb API

- ▶ **Purpose:** Fill missing values in critical columns like **rating**, **gross**, **score**, **writer**, and **votes**
- ▶ **Actions:**
 - 🔄 Used a loop to fetch real-world values via the **OMDb API**
 - 🧠 Identified rows with missing movie metadata
 - 📦 Parsed **JSON responses** for accuracy
 - 💡 Replaced nulls with verified external data instead of estimates
- ▶ **Benefits:**
 - ▶ Improves **data accuracy** without introducing artificial bias
 - ▶ Helps ensure **more reliable insights & modeling**



Stage 3 – Statistical Analysis

▶ Function used:

- ▶ `Descirbe()`, `info()`, `df.isnull.sum()`, `df.corr()`, `df['col'].value_counts()`

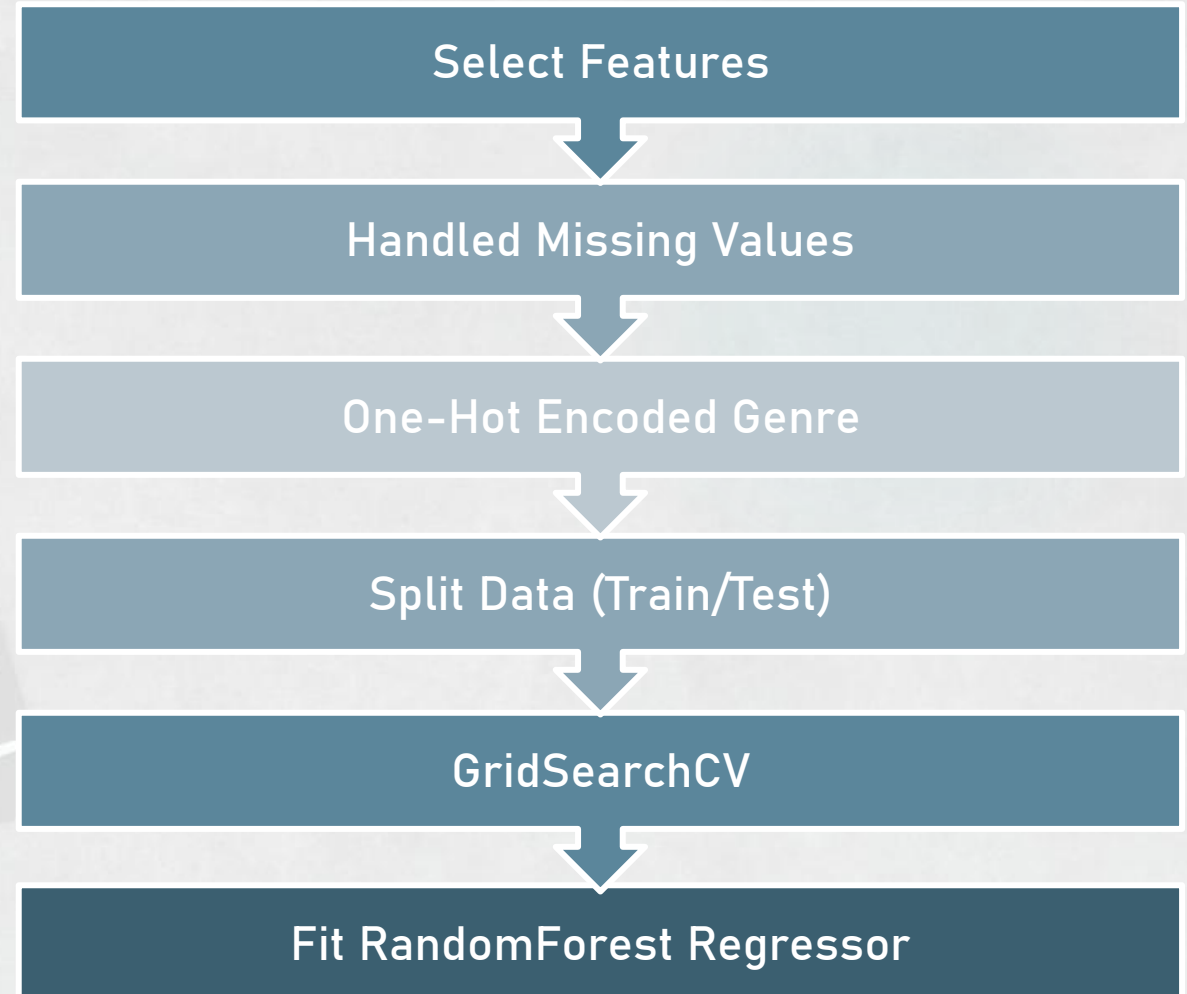
▶ Insight:

- ▶ Identified high proportion of missing values in 'budget', 'gross' column
- ▶ Verified range, central tendency, and outliers
- ▶ Prepared for model-based imputation

	year	score	votes	budget	gross	runtime	released_year
count	7668.000000	7665.000000	7.665000e+03	5.497000e+03	7.479000e+03	7664.000000	7609.000000
mean	2000.405451	6.390411	8.810850e+04	3.558988e+07	7.850054e+07	107.261613	2000.700486
std	11.153508	0.968842	1.633238e+05	4.145730e+07	1.657251e+08	18.581247	11.152091
min	1980.000000	1.900000	7.000000e+00	3.000000e+03	3.090000e+02	55.000000	1980.000000
25%	1991.000000	5.800000	9.100000e+03	1.000000e+07	4.532056e+06	95.000000	1991.000000
50%	2000.000000	6.500000	3.300000e+04	2.050000e+07	2.020576e+07	104.000000	2001.000000
75%	2010.000000	7.100000	9.300000e+04	4.500000e+07	7.601669e+07	116.000000	2010.000000
max	2020.000000	9.300000	2.400000e+06	3.560000e+08	2.847246e+09	366.000000	2020.000000

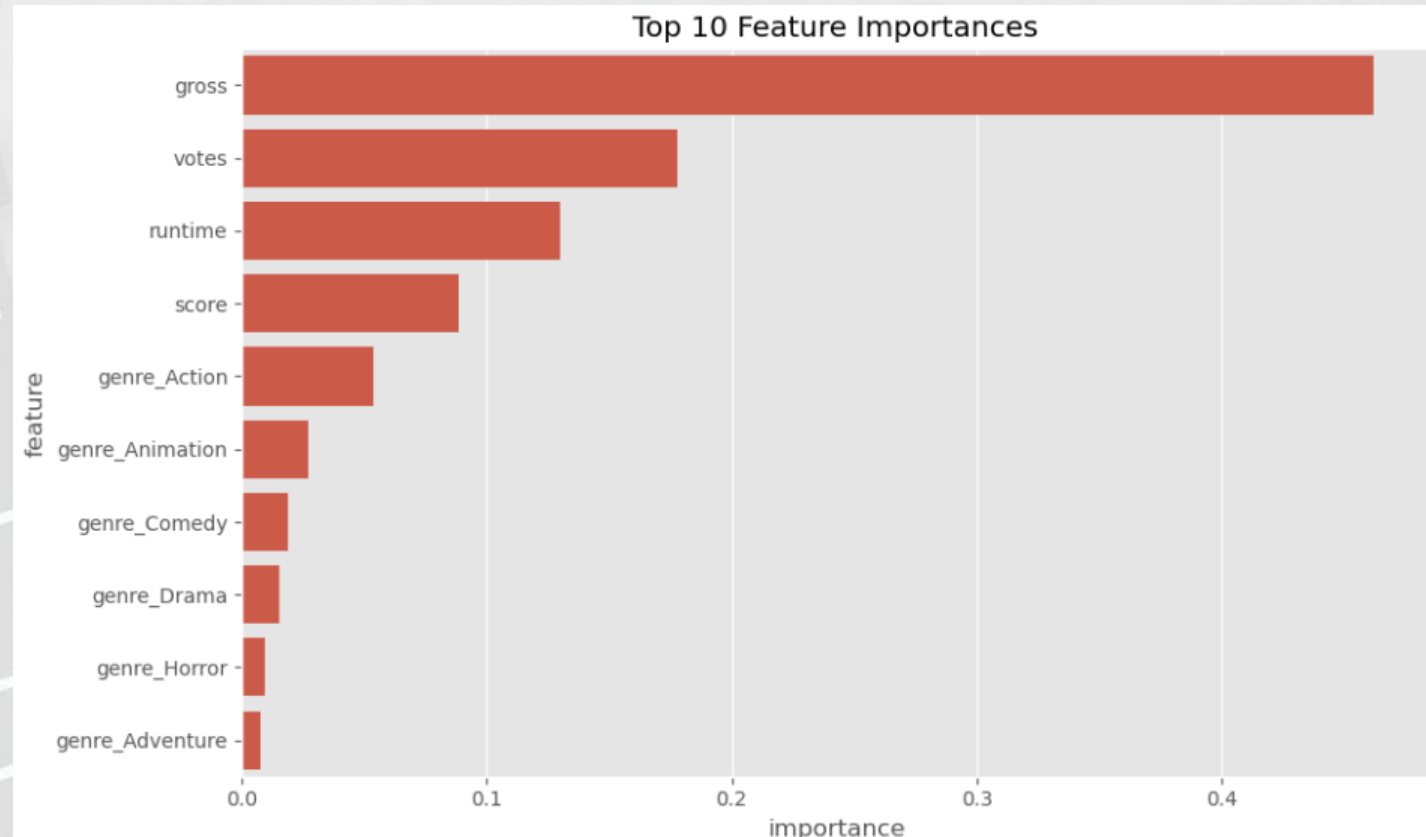
Stage 4 – Budget Regressor Model

- ▶ **Model:** RandomForestRegressor
- ▶ **Strategy:**
- ▶ **Features:** runtime, score, gross, votes and genre
- ▶ **Categorical encoding:** Genre (One-hot)
- ▶ **Null handling:** SimpleImputer with median
- ▶ **Data split:** 80% train / 20% test
- ▶ **Hyperparameter tuning:** GridSearchCV



Stage 5 – Model Evaluation

- ▶ Transformation:
 - ▶ $\log(y)$ for training → $\exp(y)$ for final prediction
- ▶ Model performance:
 - ▶ R^2 Score: 0.70
- ▶ Feature Importance:
 - ▶ Top drivers: Gross, votes, runtime, score, Genre action
- ▶ Outcome:
 - ▶ Used predicted values to fill missing budget entries



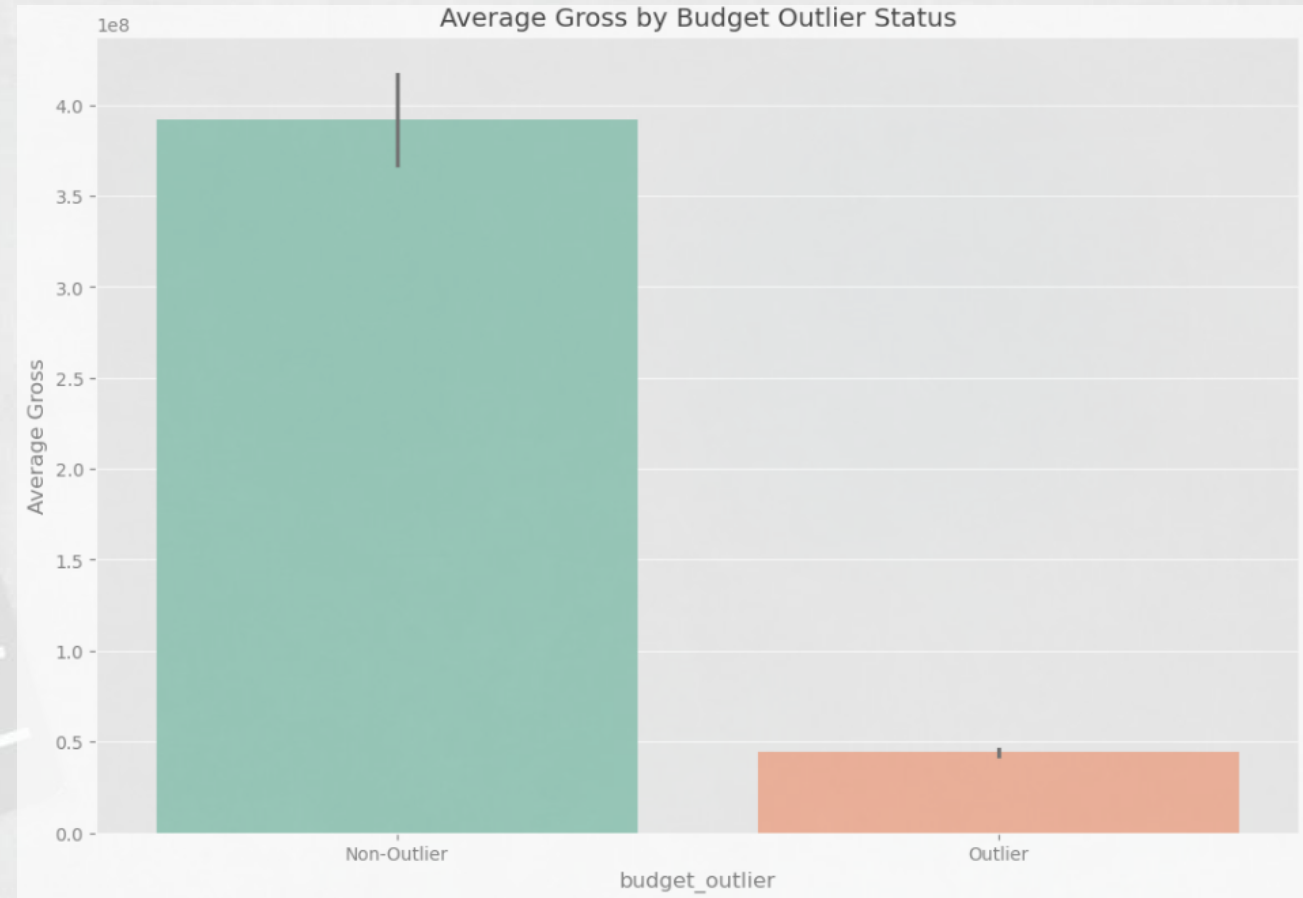
Budget Outliers & Average Gross

▶ Insights:

- ▶ Movies with extremely high or low budgets (outliers) tend to earn less on average than movies with more typical budgets

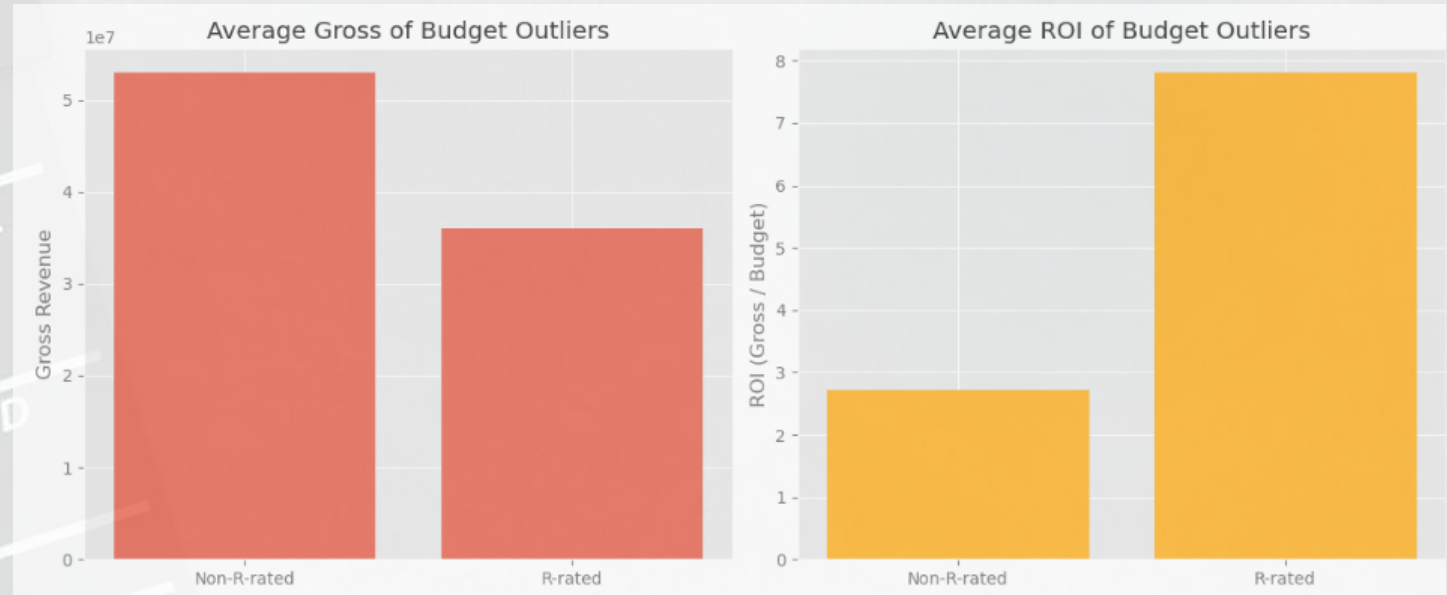
▶ Tagline:

- ▶ Bigger budget doesn't mean bigger return



R-Rated Films in Outliers

- ▶ **Insights:**
- ▶ R-rated movies dominate the budget (outlier) category, which can either flop hard or succeed wildly
- ▶ R-rated Horror has highest ROI → low budget, niche appeal, great profit (e.g., Paranormal Activity, Get Out).
- ▶ Learning: High ROI $\neq$ High Gross
- ▶ **Suggested Action:** Studios can maximize ROI by investing in mid-budget R-rated genre films

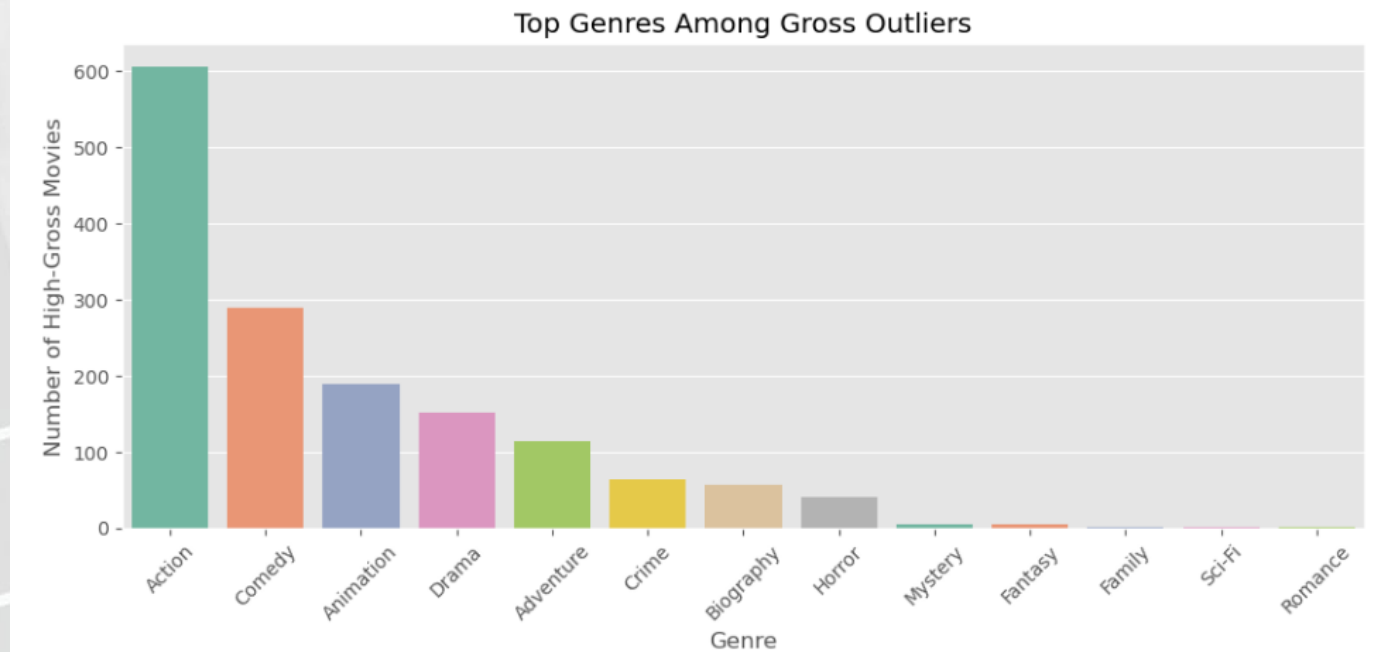


Blockbuster Insights: What \$100M+ Grossing Movies Reveal

Title	gross	budget	rating
Avatar	2.847246e+09	237000000.0	PG-13
Avengers: Endgame	2.797501e+09	356000000.0	PG-13
Titanic	2.201647e+09	200000000.0	PG-13
Star Wars: Episode VII - The Force Awakens	2.069522e+09	245000000.0	PG-13
Avengers: Infinity War	2.048360e+09	321000000.0	PG-13
The Lion King	1.670728e+09	260000000.0	PG
Jurassic World	1.670516e+09	150000000.0	PG-13
The Avengers	1.518816e+09	220000000.0	PG-13
Furious 7	1.515341e+09	190000000.0	PG-13
Frozen II	1.450027e+09	150000000.0	PG

💰 Top Earners at the Box Office

These films each grossed over \$100M, showing what peak commercial success looks like.

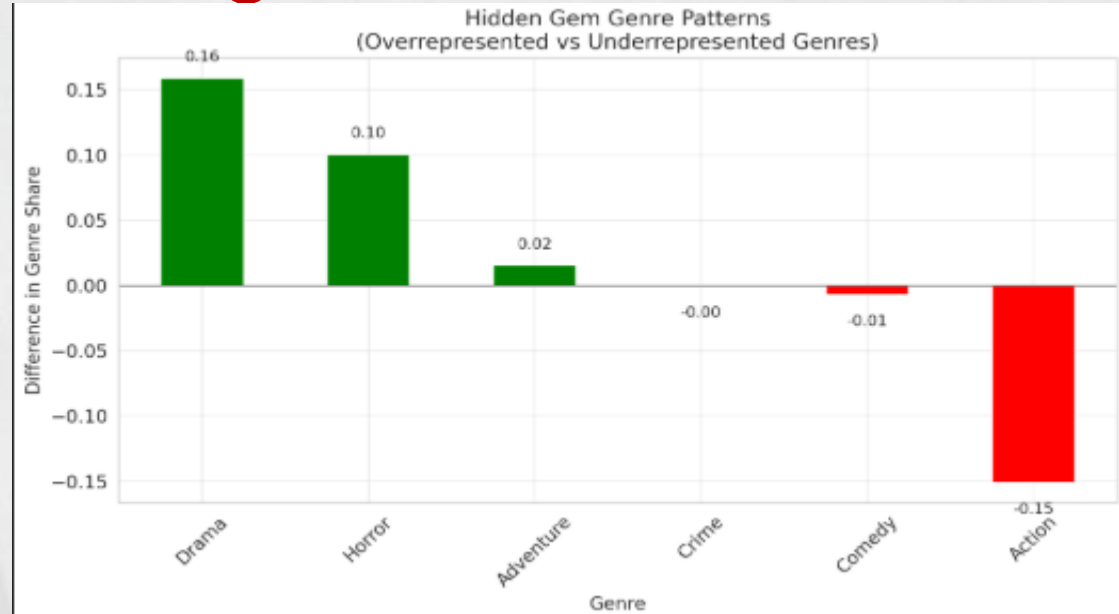


Action leads in high-grossing films, supported by blockbuster franchises and wide audience appeal. However, not all high-grossing genres deliver equally strong ROI.

**A big gross doesn't always mean big profit
— but it often starts with the right genre**

Hidden Gems: Uncovering Profitable Underdogs

Title	budget	score	gross
Airplane	3.500000e+06	7.7	83453539.0
The Terminator	6.400000e+06	8.0	78371200.0
Platoon	6.000000e+06	8.1	138545632.0
Four Weddings and a Funeral	4.400000e+06	7.1	245700832.0
The Full Monty	3.500000e+06	7.2	257938649.0
Billy Elliot	5.000000e+06	7.7	109283018.0
Lost in Translation	4.000000e+06	7.7	118686937.0
Saw	1.200000e+06	7.6	103911669.0
Crash	6.500000e+06	7.7	98410061.0
The Lives of Others	2.000000e+06	8.4	77356942.0
Get Out	4.500000e+06	7.7	255589157.0
Searching	8.800000e+05	7.6	75462037.0
The Peanut Butter Falcon	6.200000e+06	7.6	133031473.0
Open	2.007199e+06	7.9	85105259.0



🎭 **Drama** is disproportionately represented among hidden gems (+16%) — suggesting emotional stories can thrive on modest budgets.

👻 **Horror** also stands out (+10%) — aligning with previous ROI findings (cheap to make, highly profitable).

🎬 **Action films** are rare among hidden gems (−15%) — high production cost may be a limiting factor.

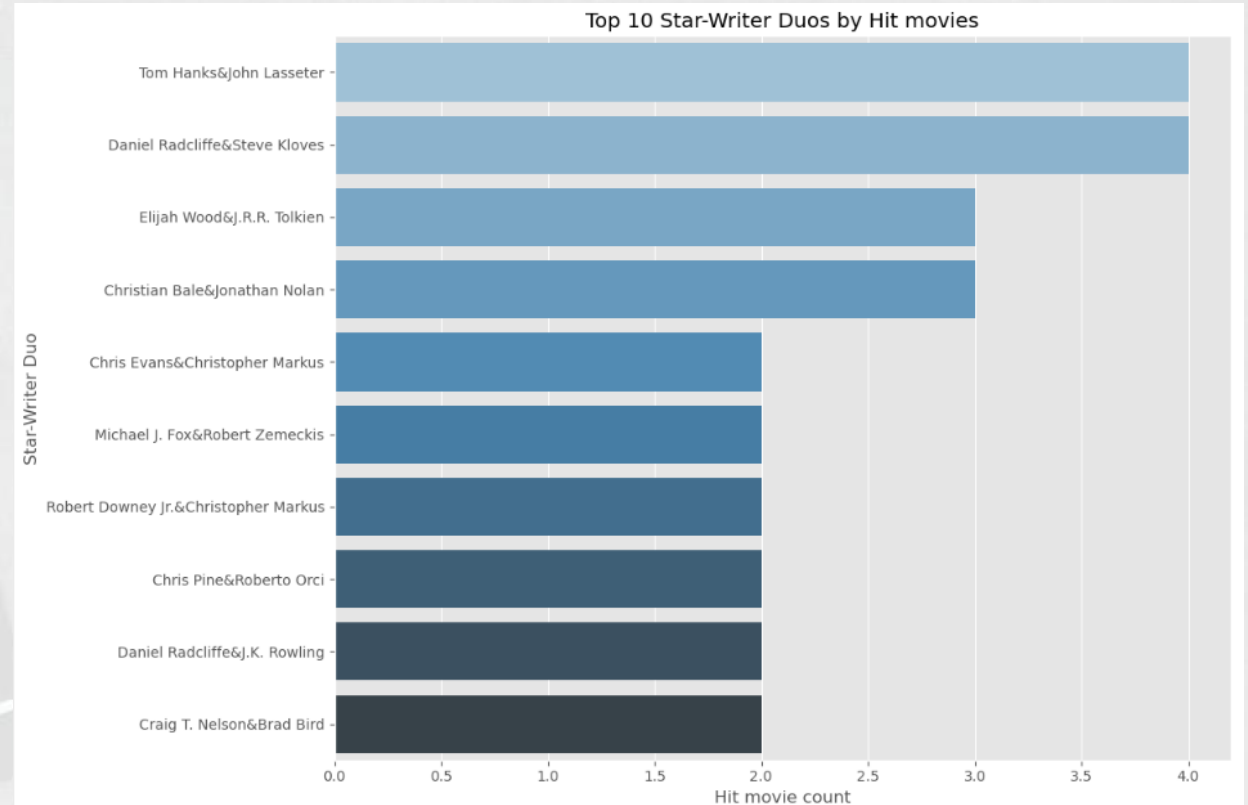
😄 **Comedy** appears consistently, but doesn't overperform in hidden gem space.

Studio ROI Rankings: Who Makes the Smartest Investments?

 Studio	Genre	 ROI	 Insight
Marvel Studios	Action	4.66	 Insanely profitable. Very few movies (12), but extremely high ROI per film.
DreamWorks Animation	Family/Animation	2.55	Strong performer in family/animated content with solid returns.
Walt Disney Pictures	Family/Animation	2.41	High ROI across 123 films — balanced between family, fantasy, and animation.
New Line Cinema	Horror	2.40	Driven by horror (The Conjuring, Annabelle) and Lord of the Rings success.
Twentieth Century Fox	Action	2.39	Solid performer with franchise power (X-Men, Avatar, etc.).
Universal Pictures	Action/Animation	2.15	High volume + franchises (Fast & Furious, Minions) = strong ROI.
Warner Bros.	Action	1.81	Big IPs but slightly lower ROI — possibly due to huge budgets (Harry Potter, DC Universe).
Columbia Pictures	Action, Drama, Family	1.71	Consistent output, but ROI may be affected by higher cost base.

Box Office Chemistry: The Power of Star–Writer Teams

- ▶ Top duo: ☒ Tied for 1st place — Tom Hanks & John Lasseter and Daniel Radcliffe & Steve Kloves (4 hit movies each)
- ▶ Most top duos are linked to long-running franchises like Toy Story, Harry Potter, Lord of the Rings
- ▶ Investing in repeat collaborations between actors and writers increases the chance of commercial success.



Strategic Takeaways

- ▶ Invest in mid-budget, R-rated genre films
- ▶ Lean into horror, family, drama for ROI
- ▶ Reuse successful star-writer pairings
- ▶ Don't equate gross with profit
- ▶ Repeat winning collaborations
- ▶  Track ROI, not just gross



Lights, Camera, Insights!

Thank you for your time and attention!

I hope this deep dive into movie data offered new perspectives on what truly drives cinematic success.

